

Linear regression

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Deterministic relation

- exact relationships
- Suitable when prediction error is negligible
- **Example:** force is exactly mass times acceleration
 - $F = m \cdot a$

Probabilistic models

- Hypothesize two components
 - Deterministic
 - Random error
- **Example:** sales volume (y) is 10 times advertising spending (x) + random error
 - $y = 10x + \varepsilon$
 - Random error may be due to factors other than advertising

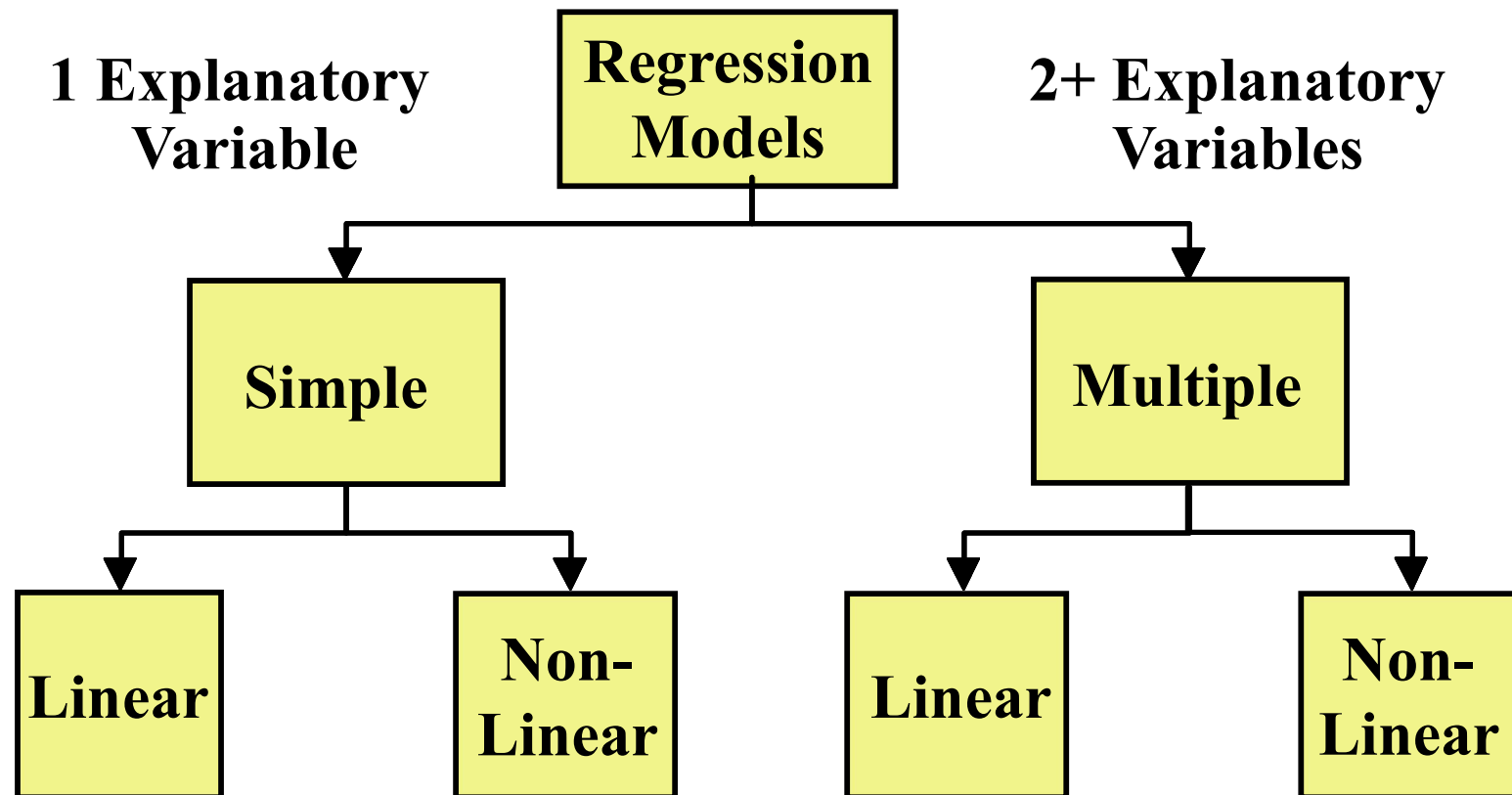
Regression models

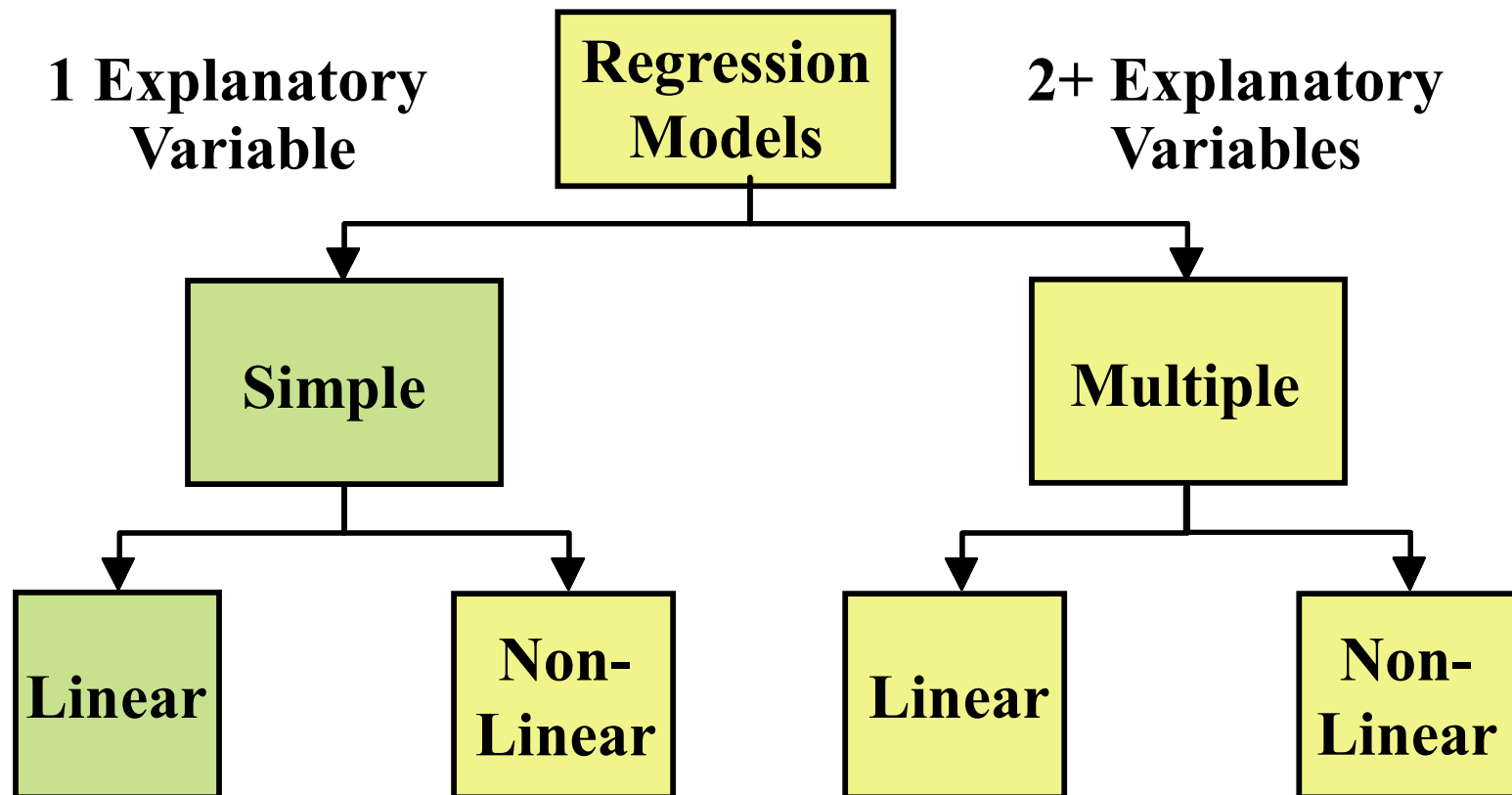
- **What is the relationship between the variables by using data?**
- **Equation used**
 - **One numerical dependent (response) variable**
 - **What is to be predicted**
 - **One or more numerical or categorical independent (explanatory) variables**

Regression Modeling Steps

1. Hypothesize deterministic component
2. Estimate unknown model parameters
3. Specify probability distribution of random error term
 - Estimate standard deviation of error
4. Evaluate model
5. Use model for prediction and estimation

Types of Regression Models





Linear Regression Model

Relationship between variables is a linear function

The diagram illustrates the Linear Regression Model equation: $y = \beta_0 + \beta_1 x + \varepsilon$. The equation is written in a dark red font. Five green arrows point from descriptive labels to the components of the equation:

- An arrow from "Dependent (Response) Variable" points to y .
- An arrow from "Population y-intercept" points to β_0 .
- An arrow from "Population Slope" points to β_1 .
- An arrow from "Independent (Explanatory) Variable" points to x .
- An arrow from "Random Error" points to ε .

Population y-intercept

Population Slope

Random Error

Dependent (Response) Variable

Independent (Explanatory) Variable

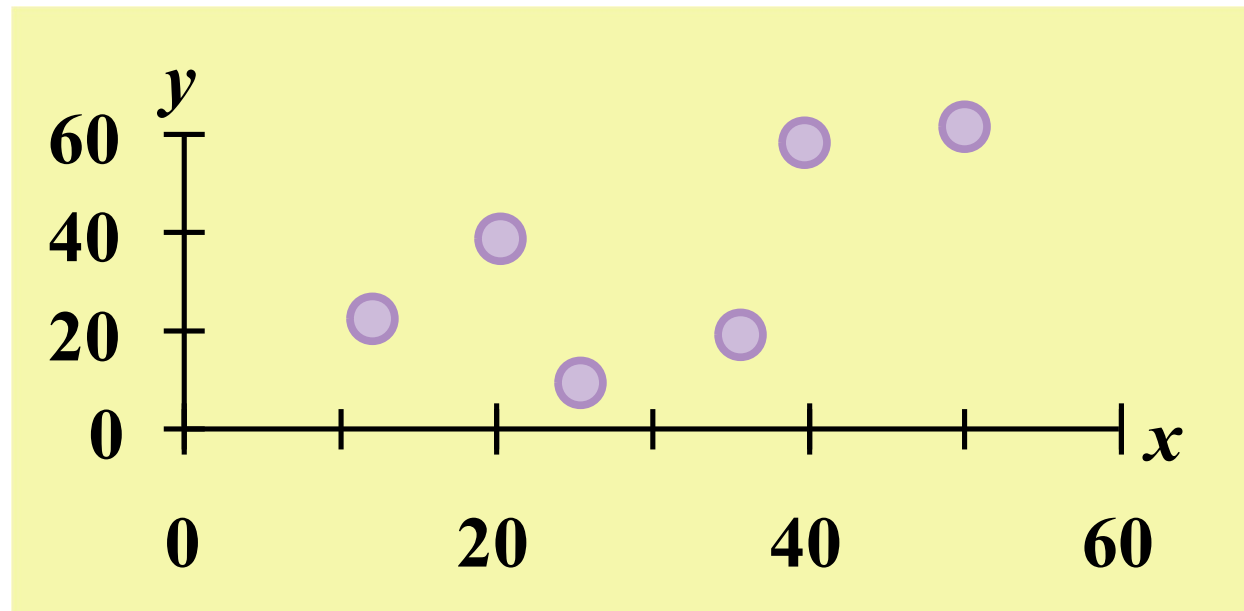
$$y = \beta_0 + \beta_1 x + \varepsilon$$

Regression Modeling Steps

1. Hypothesize deterministic component
2. **Estimate unknown model parameters**
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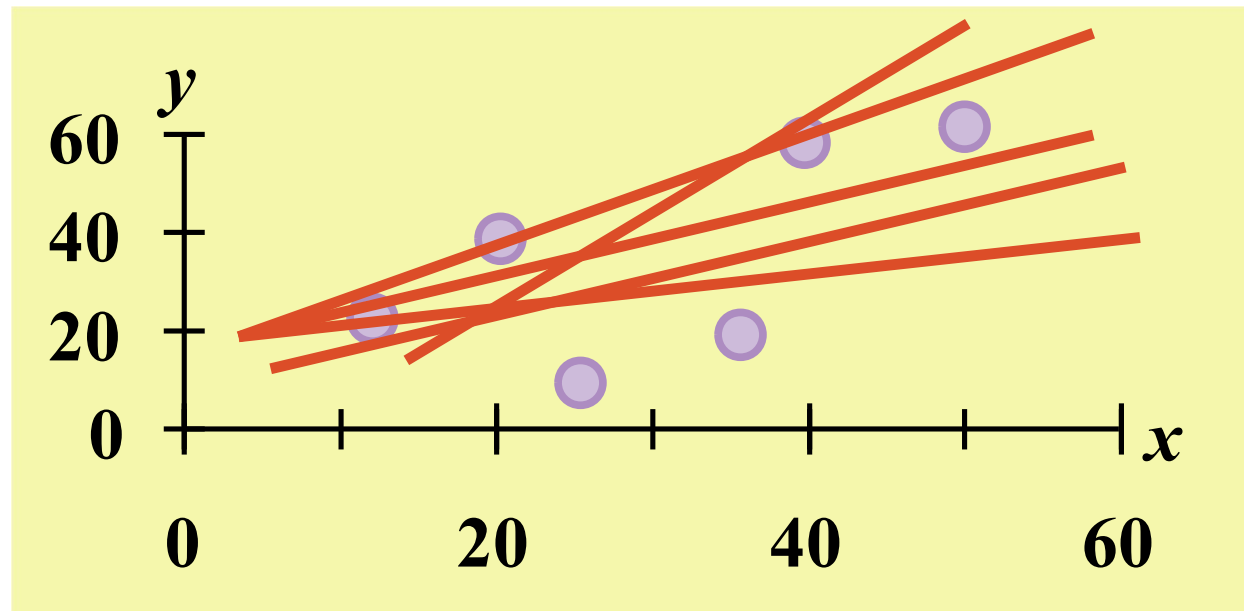
Scattergram

1. Plot of all (x_i, y_i) pairs
2. Suggests how well model will fit



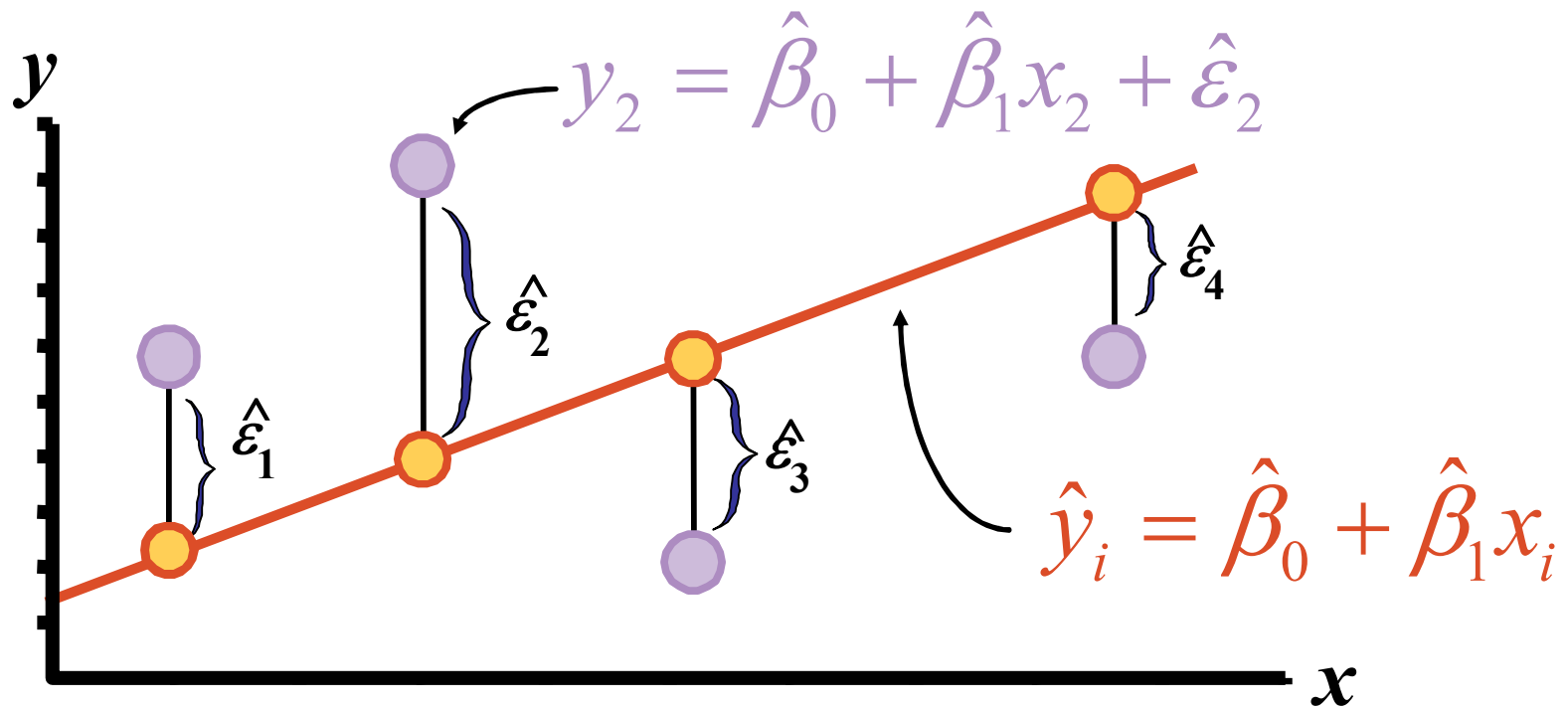
Thinking Challenge

- How would you draw a line through the points?
- How do you determine which line 'fits best'?



Least Squares

LS minimizes $\sum_{i=1}^n \hat{\varepsilon}_i^2 = \hat{\varepsilon}_1^2 + \hat{\varepsilon}_2^2 + \hat{\varepsilon}_3^2 + \hat{\varepsilon}_4^2$



Coefficient Equations

Prediction Equation $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$

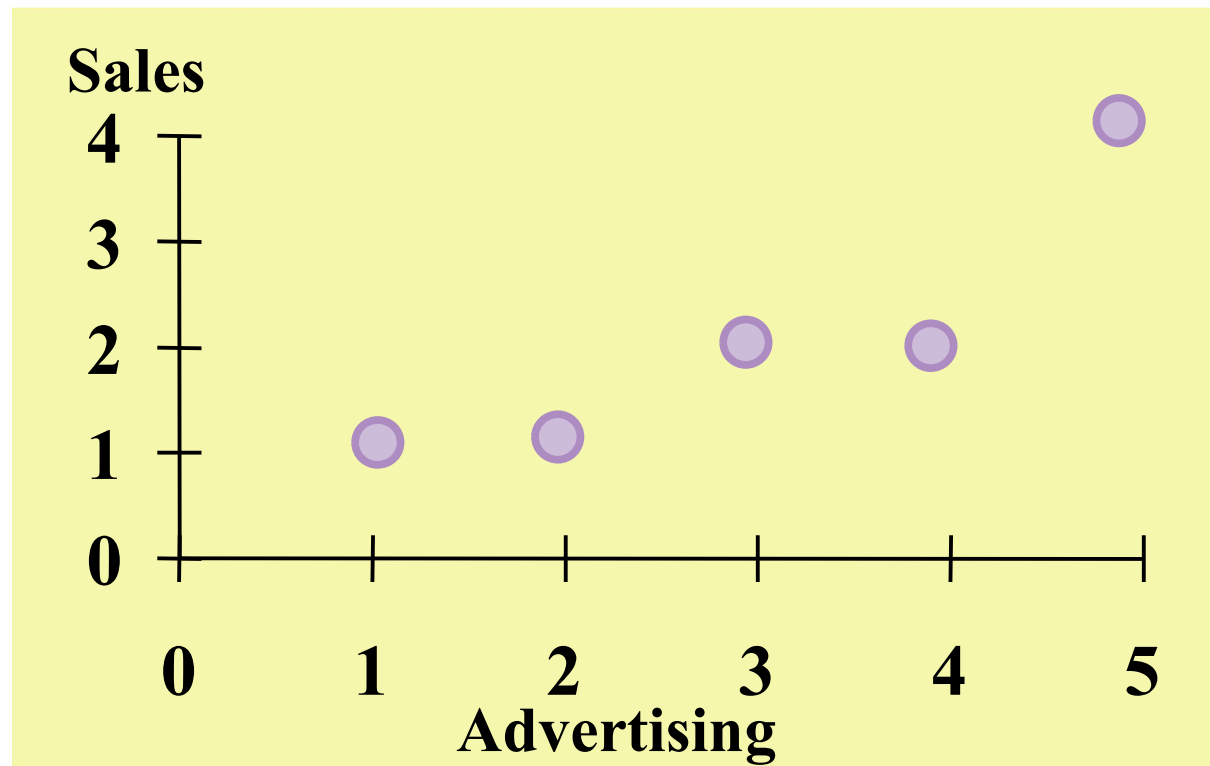
Slope
$$\hat{\beta}_1 = \frac{SS_{xy}}{SS_{xx}} = \frac{\sum_{i=1}^n x_i y_i - \frac{\left(\sum_{i=1}^n x_i\right)\left(\sum_{i=1}^n y_i\right)}{n}}{\sum_{i=1}^n x_i^2 - \frac{\left(\sum_{i=1}^n x_i\right)^2}{n}}$$

y-intercept $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$

Computation Table

x_i	y_i	x_i^2	y_i^2	$x_i y_i$
x_1	y_1	x_1^2	y_1^2	$x_1 y_1$
x_2	y_2	x_2^2	y_2^2	$x_2 y_2$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
x_n	y_n	x_n^2	y_n^2	$x_n y_n$
Σx_i	Σy_i	Σx_i^2	Σy_i^2	$\Sigma x_i y_i$

Sales vs. Advertising



x_i	y_i	x_i^2	y_i^2	$x_i y_i$
1	1	1	1	1
2	1	4	1	2
3	2	9	4	6
4	2	16	4	8
5	4	25	16	20
15	10	55	26	37

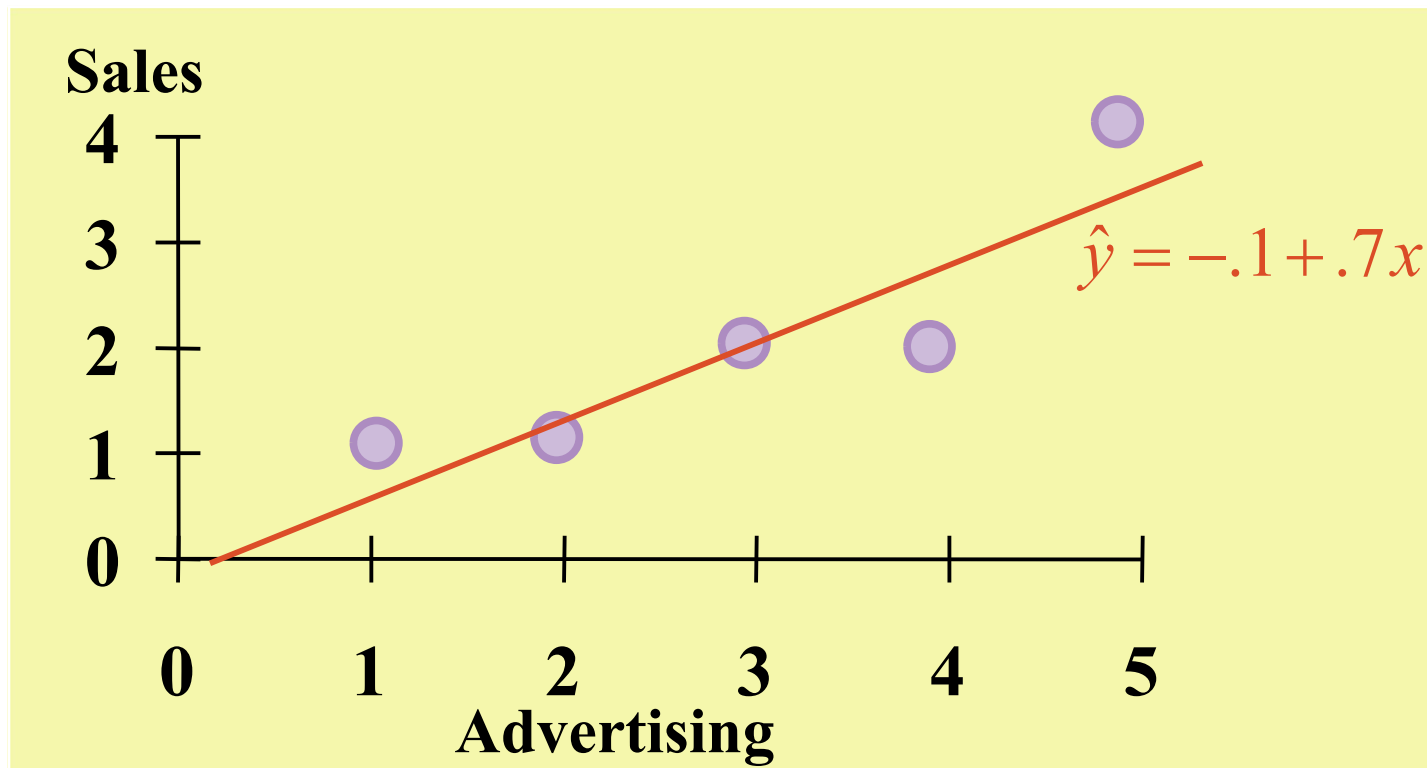
Parameter Estimation Solution

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n x_i y_i - \frac{\left(\sum_{i=1}^n x_i\right)\left(\sum_{i=1}^n y_i\right)}{n}}{\sum_{i=1}^n x_i^2 - \frac{\left(\sum_{i=1}^n x_i\right)^2}{n}} = \frac{37 - \frac{(15)(10)}{5}}{55 - \frac{(15)^2}{5}} = .70$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 2 - (.70)(3) = -.10$$

$$\hat{y} = -.1 + .7x$$

Regression Line Fitted to the Data



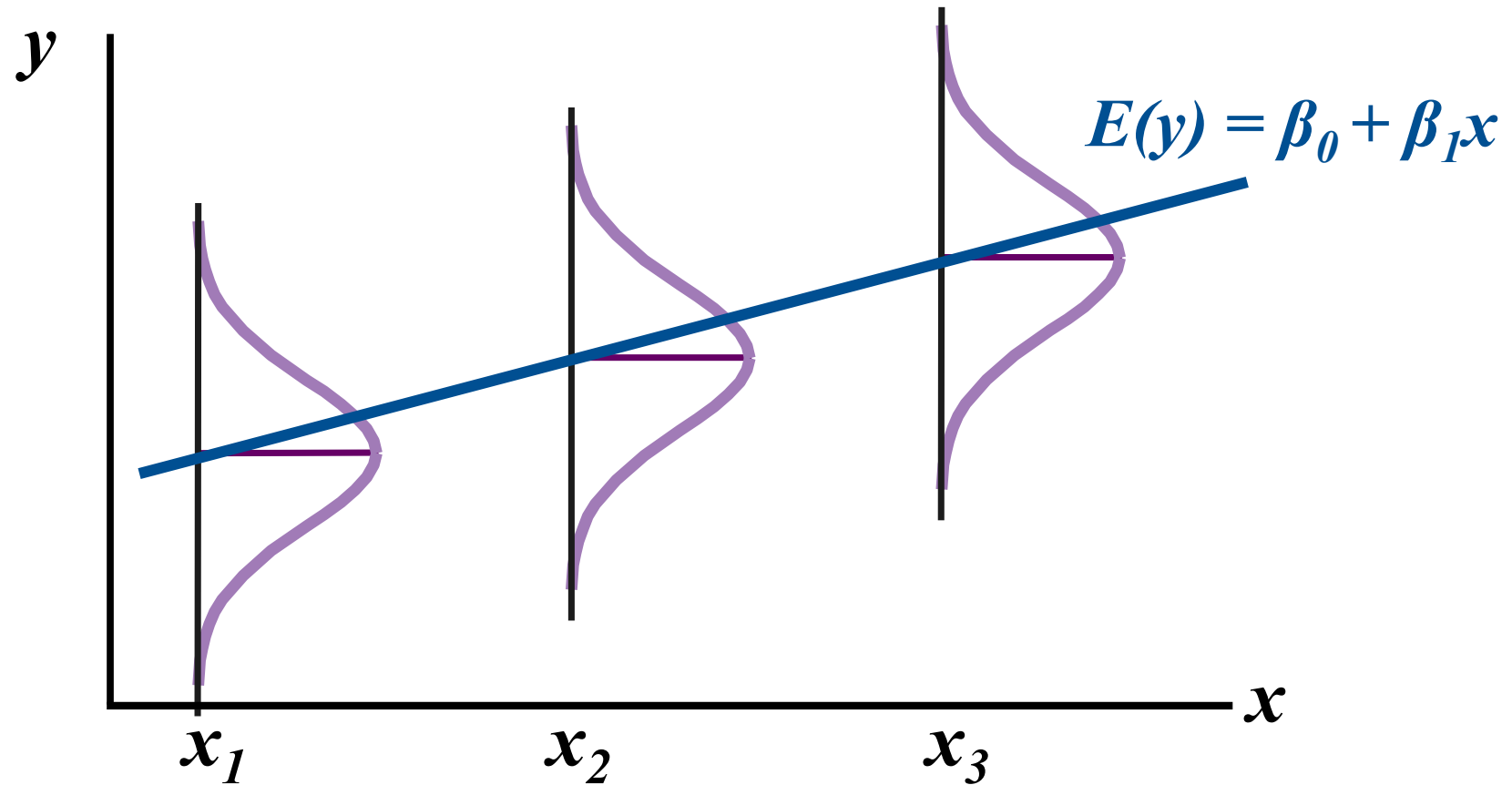
Regression Modeling Steps

1. Hypothesize deterministic component
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Linear Regression Assumptions

- Mean of probability distribution of error, ε , is 0
- Probability distribution of error has constant variance σ^2
- Probability distribution of error is normal
- Errors are independent

Error Probability Distribution



Estimation of σ^2

$$s^2 = \frac{SSE}{n-2} \quad \text{where} \quad SSE = \sum (y_i - \hat{y}_i)^2$$

$$s = \sqrt{s^2} = \sqrt{\frac{SSE}{n-2}}$$

Calculating SSE Solution

x_i	y_i	$\hat{y} = -.1 + .7x$	$y - \hat{y}$	$(y - \hat{y})^2$
1	1	.6	.4	.16
2	1	1.3	-.3	.09
3	2	2	0	0
4	2	2.7	-.7	.49
5	4	3.4	.6	.36
				SSE=1.1

Calculating s^2 and s

Solution

$$s^2 = \frac{SSE}{n-2} = \frac{1.1}{5-2} = .36667$$

$$s = \sqrt{.36667} = .6055$$

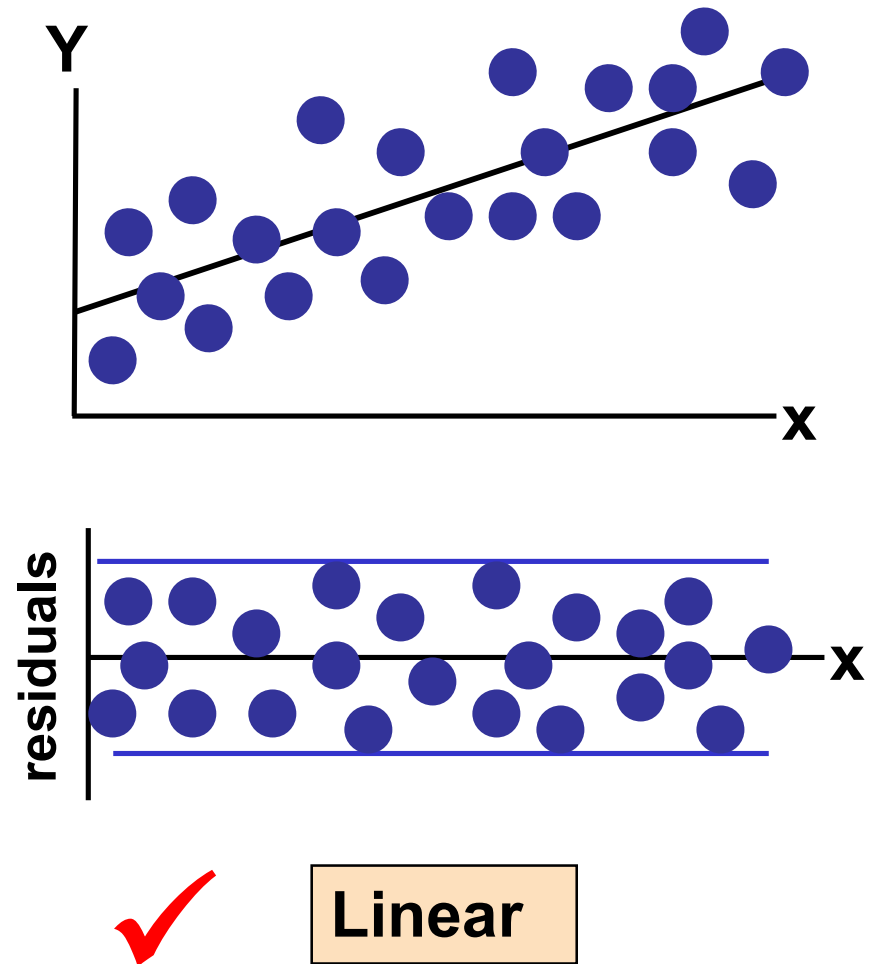
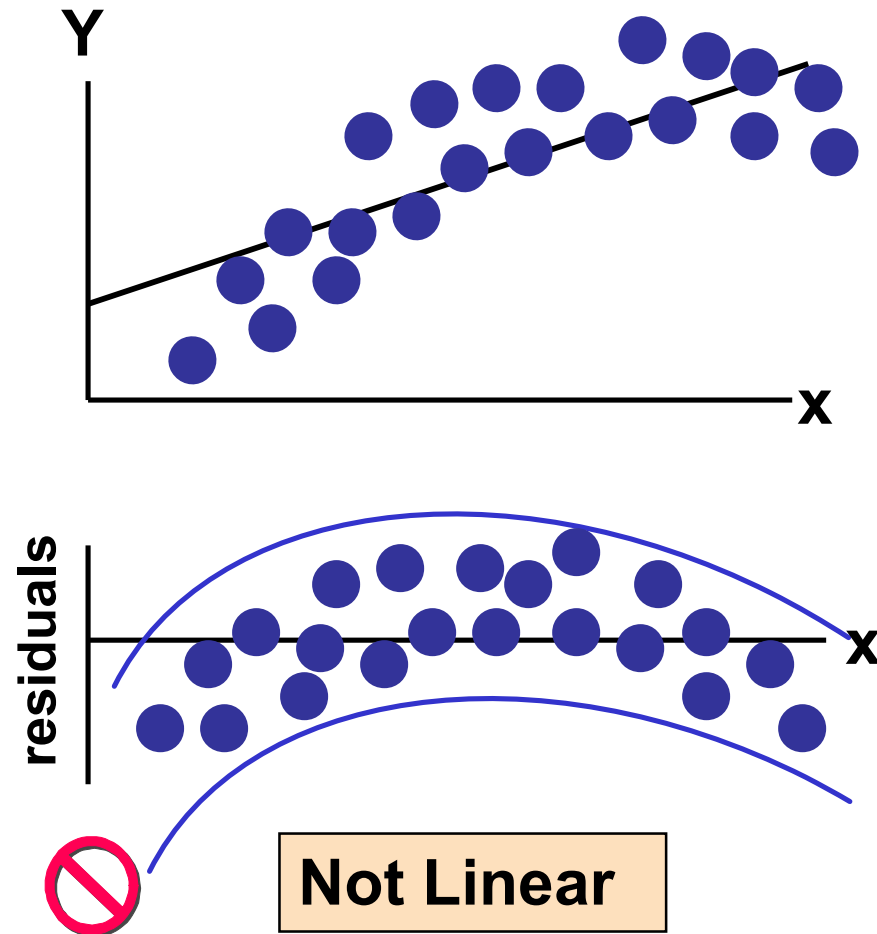
Residual Analysis

- The residual for observation i is the difference between its observed and predicted value

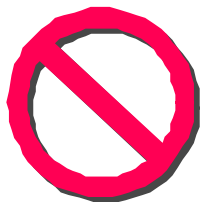
$$e_i = Y_i - \hat{Y}_i$$

- Check the assumptions of regression by examining the residuals
 - Examine for linearity assumption
 - Evaluate independence assumption
 - Examine for constant variance for all levels of X (homoscedasticity)
 - Evaluate normal distribution assumption

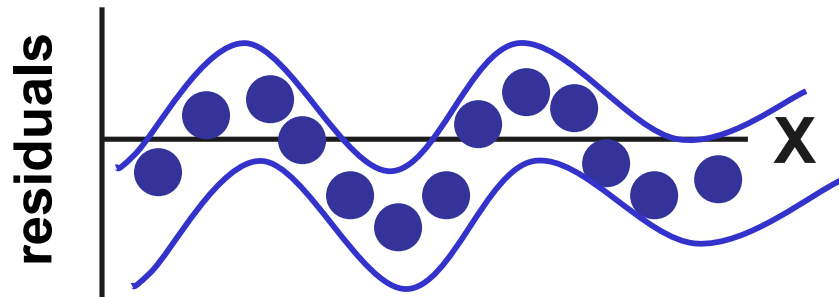
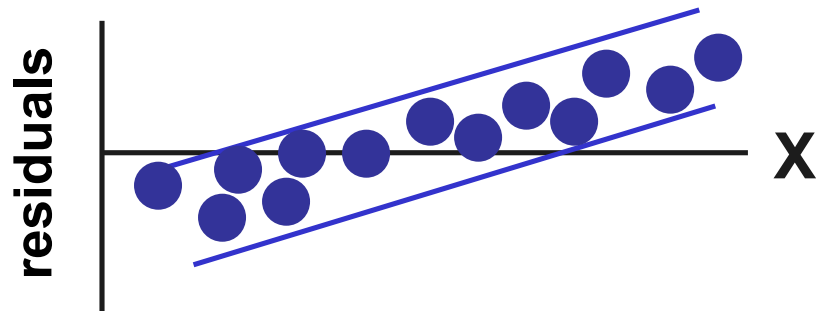
Residual Analysis for Linearity



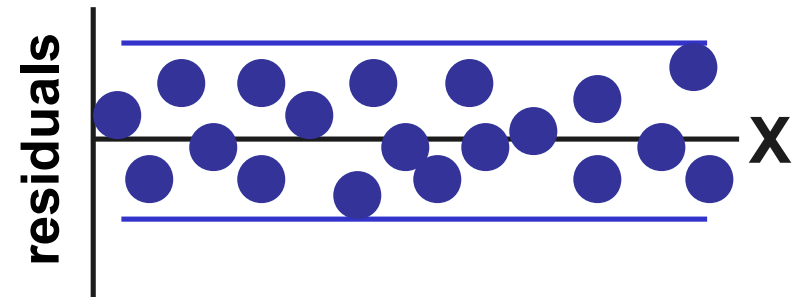
Residual Analysis for Independence



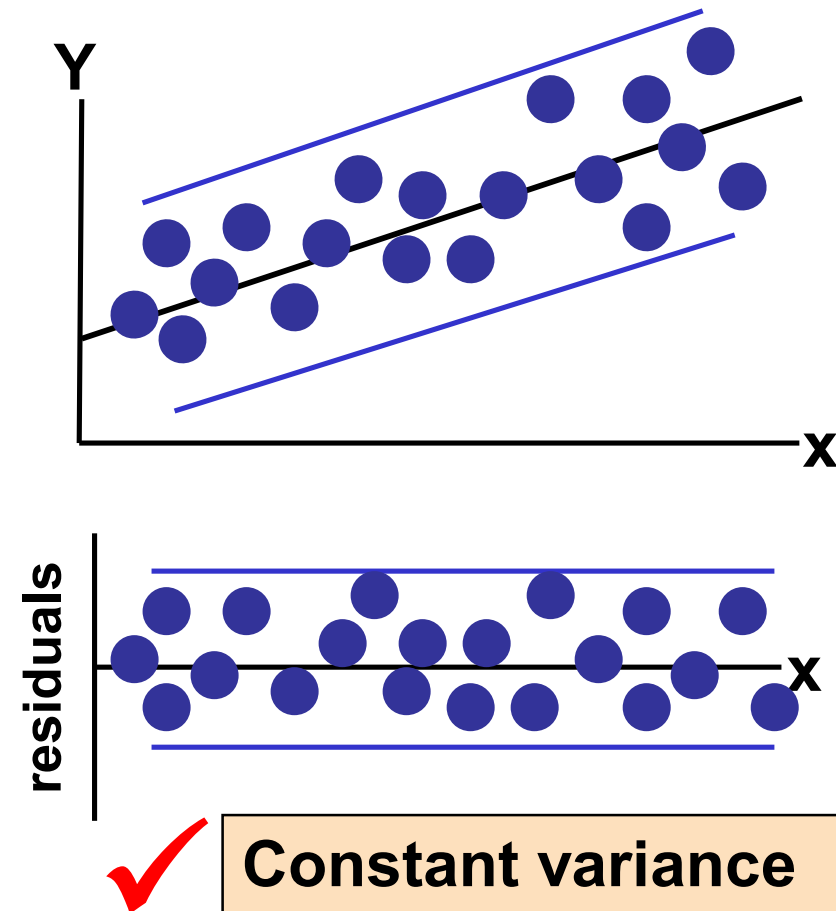
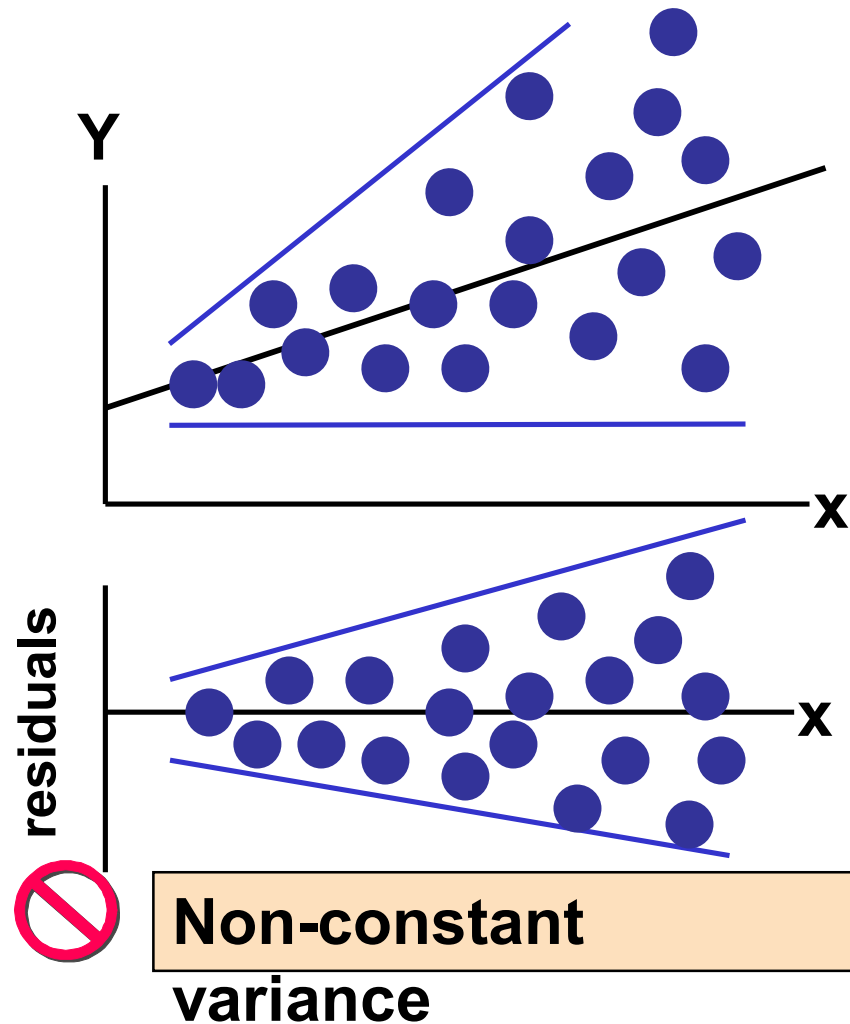
Not Independent



Independent



Residual Analysis for Equal Variance



Checking for Normality

- QQ plot
- PP plot
- Non-parametric test

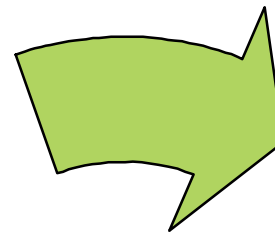
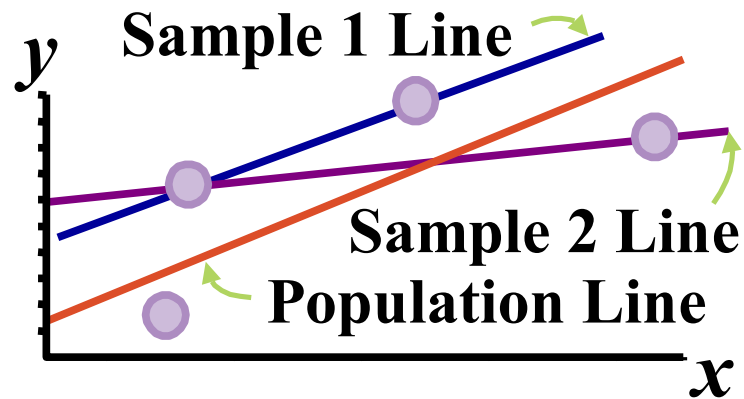
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Test of Slope Coefficient

- Shows if there is a linear relationship between x and y
- Hypotheses
 - $H_0: \beta_1 = 0$ (No Linear Relationship)
 - $H_a: \beta_1 \neq 0$ (Linear Relationship)

Sampling Distribution of Sample Slopes

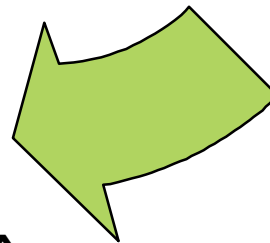
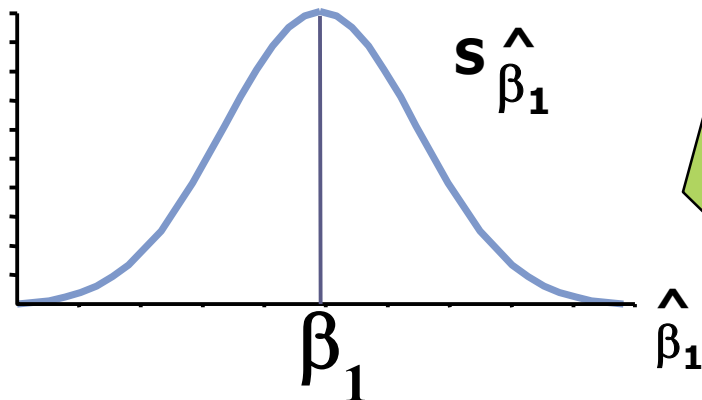


All Possible
Sample Slopes

Sample 1:	2.5
Sample 2:	1.6
Sample 3:	1.8
Sample 4:	2.1
:	:

Very large number of
sample slopes

Sampling Distribution



Slope Coefficient Test Statistic

$$t = \frac{\hat{\beta}_1}{S_{\hat{\beta}_1}} = \frac{\hat{\beta}_1}{s / \sqrt{SS_{xx}}} \quad df = n - 2$$

where

$$SS_{xx} = \sum_{i=1}^n x_i^2 - \frac{\left(\sum_{i=1}^n x_i \right)^2}{n}$$

Solution Table

x_i	y_i	x_i^2	y_i^2	$x_i y_i$
1	1	1	1	1
2	1	4	1	2
3	2	9	4	6
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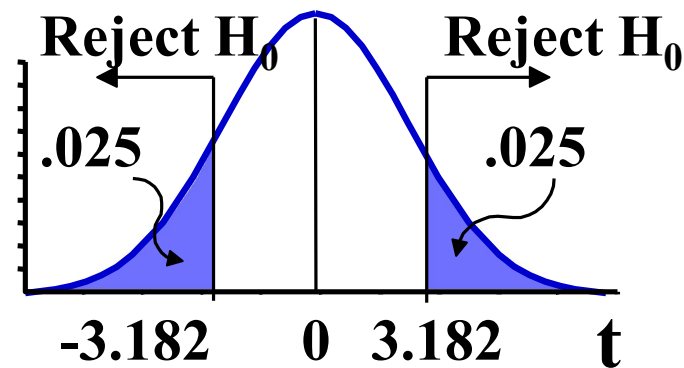
Test Statistic Solution

$$S_{\hat{\beta}_1} = \frac{S}{\sqrt{SS_{xx}}} = \frac{.6055}{\sqrt{55 - \frac{(15)^2}{5}}} = .1914$$

$$t = \frac{\hat{\beta}_1}{S_{\hat{\beta}_1}} = \frac{.70}{.1914} = 3.657$$

Test of Slope Coefficient Solution

- **$H_0: \beta_1 = 0$**
- **$H_a: \beta_1 \neq 0$**
- **$\alpha = .05$**
- **$df = 5 - 2 = 3$**
- **Critical Value(s):**



Test of Slope Coefficient Solution

Test Statistic:

$$t = \frac{\hat{\beta}_1}{S_{\hat{\beta}_1}} = \frac{.70}{.1914} = 3.657$$

Decision:

Reject at $\alpha = .05$

Conclusion:

There is evidence of a relationship

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Prediction With Regression Models

- Types of predictions
 - Point estimates
 - Interval estimates
- What is predicted
 - Population mean response $E(y)$ for given x
 - Point on population regression line
 - Individual response (y_i) for given x

Confidence Interval Estimate for Mean Value of y at $x = x_p$

$$\hat{y} \pm t_{\alpha/2} S \sqrt{\frac{1}{n} + \frac{(x_p - \bar{x})^2}{SS_{xx}}}$$

$$\mathbf{df} = n - 2$$

Solution Table

x_i	y_i	x_i^2	y_i^2	$x_i y_i$
1	1	1	1	1
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3	2	9	4	6
4	2	16	4	8
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Confidence Interval Estimate Solution

$$\hat{y} \pm t_{\alpha/2} s \sqrt{\frac{1}{n} + \frac{(x_p - \bar{x})^2}{SS_{xx}}}$$

x to be predicted

$$\hat{y} = -.1 + (.7)(4) = 2.7$$

$$2.7 \pm (3.182)(.6055) \sqrt{\frac{1}{5} + \frac{(4-3)^2}{10}}$$

$$1.645 \leq E(Y) \leq 3.755$$

Prediction Interval of Individual Value of y at $x =$ x_p

$$\hat{y} \pm t_{\alpha/2} S \sqrt{1 + \frac{1}{n} + \frac{(x_p - \bar{x})^2}{SS_{xx}}}$$

Note!

$$df = n - 2$$

Solution Table

x_i	y_i	x_i^2	y_i^2	$x_i y_i$
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2	1	4	1	2
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Prediction Interval Solution

$$\hat{y} \pm t_{\alpha/2} s \sqrt{1 + \frac{1}{n} + \frac{(x_p - \bar{x})^2}{SS_{xx}}}$$

x to be predicted

$$\hat{y} = -.1 + (.7)(4) = 2.7$$

$$2.7 \pm (3.182)(.6055) \sqrt{1 + \frac{1}{5} + \frac{(4-3)^2}{10}}$$

$$.503 \leq y_4 \leq 4.897$$

Coefficient of determination (可决系数、判定系数)

$$R^2 = \frac{SS_{xy}}{SS_{xx} SS_{yy}}$$